

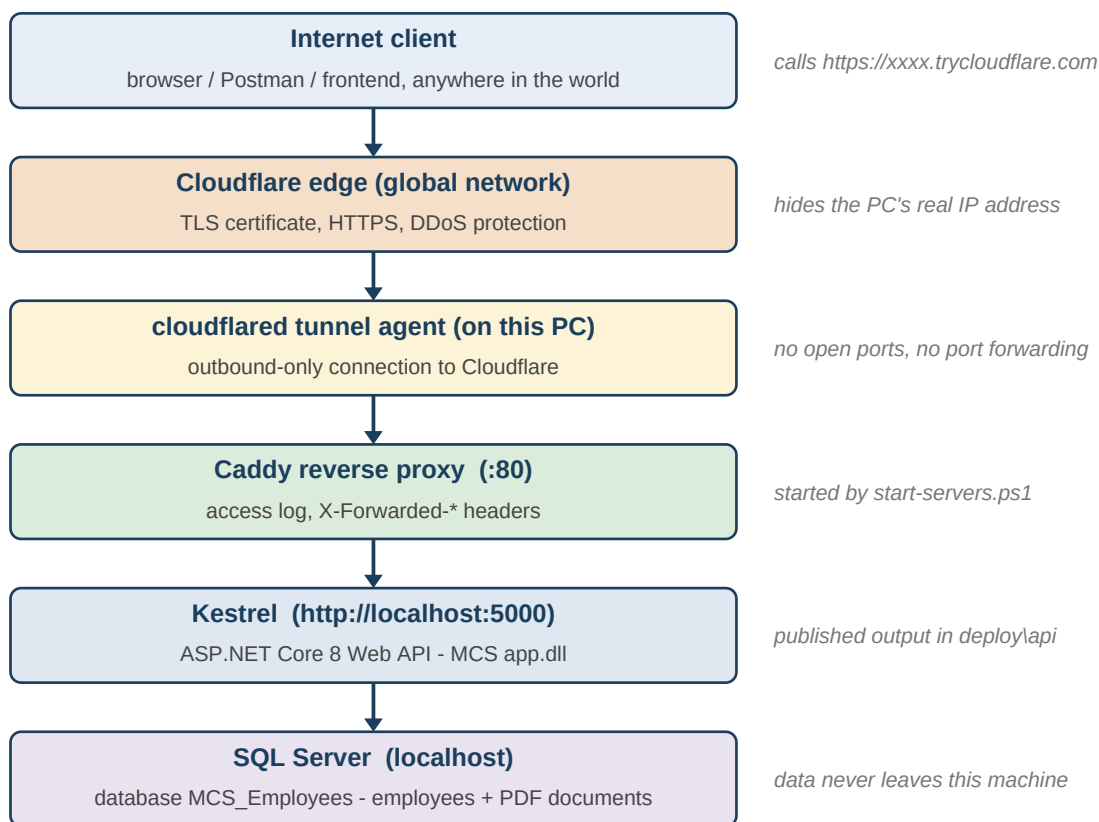
Cloudflare as a Layer Above Caddy + Kestrel

MCS Employee API | July 14, 2026

How it works

The API is exposed to the internet through a **Cloudflare quick tunnel**. The **cloudflared** agent on the PC opens an **outbound-only** connection to Cloudflare's global network and receives a random public URL (`https://xxxx.trycloudflare.com`). Internet requests hit Cloudflare's edge over HTTPS, travel down the tunnel to the PC, and are handed to Caddy, which forwards them to Kestrel exactly as before. Because the connection is outbound-only, **no ports are opened** on the PC, the machine's **IP address stays hidden**, and only `localhost:80` (Caddy → the API) is reachable - nothing else on the machine is exposed.

Layer diagram



Steps to start the backend

```
cd "D:\github projects\MCS intern"
```

```
.\start-servers.ps1      # 1. start Kestrel (localhost:5000) + Caddy (:80)
.\verify-database.ps1    # 2. (optional) confirm stack + database are healthy
.\start-tunnel.ps1       # 3. (optional) expose to the internet; prints the
                        #    https://xxxx.trycloudflare.com public URL

.\stop-servers.ps1       # stops Kestrel, Caddy and the tunnel
```

Local access works without the tunnel: `http://localhost/swagger` and `http://localhost/api/employees`. The tunnel is only needed for access from outside the machine/network.

Security notes

- Cloudflare provides the TLS certificate (HTTPS) and DDoS protection at its edge - free plan.
- The public URL is random and **changes on every tunnel restart**; suitable for demos.
- The API has **no authentication**, so stop the tunnel when finished (Ctrl+C or `stop-servers.ps1`).
- For a permanent URL and dashboard security features (WAF rules, Cloudflare Access login), add a domain to the Cloudflare account and switch to a named tunnel.